Manu Nicholas Jacob

Hardware Engineer, Root Cause Engineering · Dell Technologies · Austin, Texas
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Hardware engineer and independent researcher. Board-level failure analysis on enterprise AI server platforms, and 14 manuscripts, 11 under peer review at IEEE, ACM and Elsevier venues, on inference performance, power and memory behaviour.

WORK EXPERIENCE

Hardware Engineer · **Dell Technologies, Root Cause Engineering, Austin, TX** Jun 2025 to present

Hardware engineer on enterprise AI server platforms: system-level and board-level debug, failure analysis and root-cause engineering across PCIe, memory, power and GPU subsystems, isolating faults by controlled reproduction rather than component swapping.

- Built an **agent-based** failure-analysis tracking system linking quality to engineering, centralizing case ownership and status and shortening turnaround in the rapid-response workflow.
- **Named inventor** on a Dell Technologies invention disclosure authorized for U.S. patent filing.
- On a **performance-analysis rotation** within the role (Feb to Apr 2026), built an automated server benchmarking and power-telemetry platform that replaced a 4-hour manual workflow with a 100-minute run, **62% faster** and 5 minutes of operator time.
- Diagnosed a riser detection failure down to a single open boost-converter inductor using an oscilloscope, logic analyzer and micro X-ray imaging, proving mechanical solder-joint separation rather than electrical overstress, and separated genuine component failures from system-level interactions across recurring controller errors and repeat no-boot cases.

Root Cause Engineering Intern · **Dell Technologies, Austin, TX** May to Aug 2024

- Developed a unified diagnostic tool for NVIDIA AI platforms that detected PCIe replays, bus resets and GPU faults through a single workflow.
- Implemented optimized secondary-bus-reset procedures and real-time PCIe replay detection, cutting link retrain time and improving fault isolation on the debug bench.

Controls Engineering Intern · **Schneider Electric, Lexington, KY** Jun to Aug 2023

- Designed and deployed a human-machine interface for palletizer control on the factory floor and optimized PLC-to-HMI communication on the packer line, improving line efficiency and reducing downtime.
- Led an overhaul of internal standardization specifications against ANSI/RIA and OSHA requirements, adding a dedicated section for robotic cells.

SKILLS

Hardware: enterprise servers, PCIe, GPU subsystems, DDR memory, power delivery, board-level debug, oscilloscope, logic analyzer, X-ray inspection, embedded systems, microcontrollers, custom PCB.

Software: Python, C, C++, SQL, Bash, Linux, PyTorch, llama.cpp, Linux `perf`, hardware performance counters.

Infrastructure: Kubernetes (k3s), Docker, cgroups, Prometheus, Grafana, distributed job scheduling.

Certifications: 3 Oracle Cloud Infrastructure 2025 certifications: Generative AI Professional, AI Foundations, Foundations.

Coursework: NVIDIA AI Infrastructure and Operations Fundamentals (course, 2026).

EDUCATION

University of Massachusetts Amherst · **B.S. Electrical Engineering and B.S. Computer Engineering** 2021 to May 2025

RESEARCH AND PUBLICATIONS

- **14 manuscripts, 11 under peer review** at IEEE Transactions on Computers, IEEE TPDS, ACM TECS, IEEE Internet of Things Journal, IEEE Embedded Systems Letters, IEEE Design & Test, Elsevier SUSCOM and HotMobile 2027.
- 4 TechRxiv preprints; every result ships with a DOI-archived artifact. manunicholasjacob.com/research.

OPEN SOURCE AND PROFESSIONAL SERVICE

Merged 16 pull requests into other projects, 4 into **ai-dynamo AIPerf**, 3 into **NVIDIA garak** and an **Arm Learning Path**, with 15 more open there and at **vLLM**; published **llama-roofline** and **ML Systems Lab** to PyPI, plus **15 DOI-archived artifacts** on Zenodo. **Artifact Evaluation Committee, ATC '26** (ACM SIGOPS, formerly USENIX ATC; CORE A); **reviewer, Journal of Open Source Software** (four reviews completed, one published; two in progress); **Judge, HackTX 2025** (UT Austin, 700+ participants).